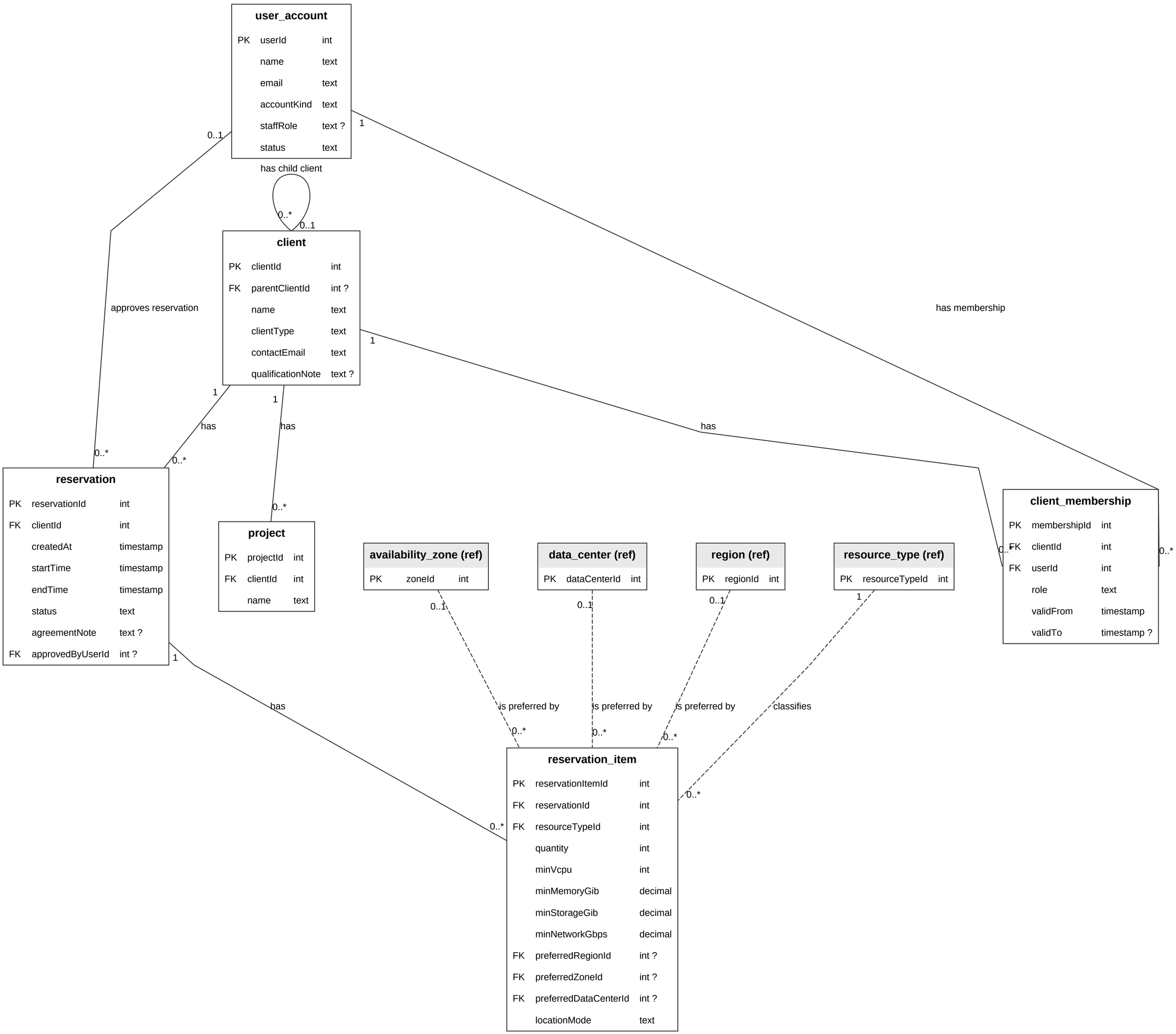
Clients and reservations - ER diagram

Member A: 6 tables; shaded boxes reference another domain

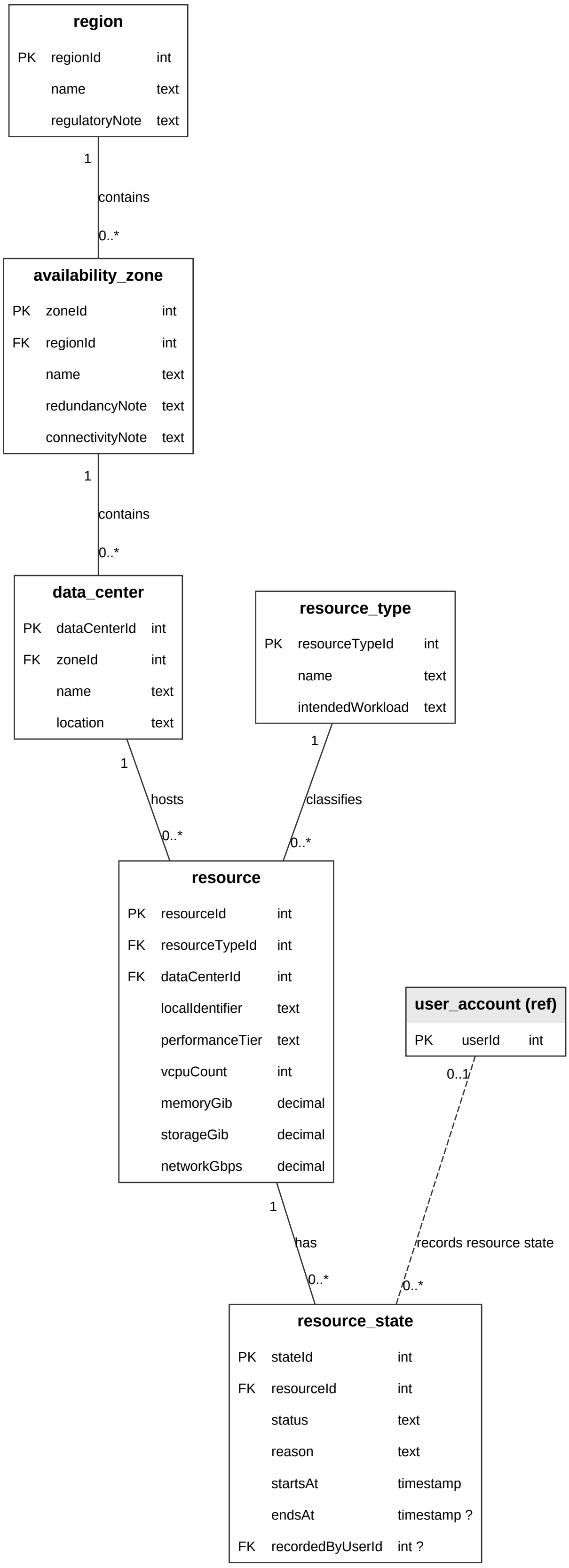


Assumptions and notation

PK = primary key; FK = foreign key; ? = nullable. Unmarked attributes are required.
Endpoint cardinalities: 1 = exactly one; 0..1 = zero or one; 0..* = zero or more.
Each labelled line represents its child foreign key. Reference boxes repeat existing entities.
Uniqueness, time intervals, allocation limits and billing rules: analysis report R1-R9.
A client has at most one parent; client hierarchy cycles are prohibited.
An item selects at most one preferred location level; approved reservations must contain items.

Infrastructure and availability - ER diagram

Member B: 6 tables; shaded boxes reference another domain



Assumptions and notation

PK = primary key; FK = foreign key; ? = nullable. Unmarked attributes are required.

Endpoint cardinalities: 1 = exactly one; 0..1 = zero or one; 0..* = zero or more.

Each labelled line represents its child foreign key. Reference boxes repeat existing entities.

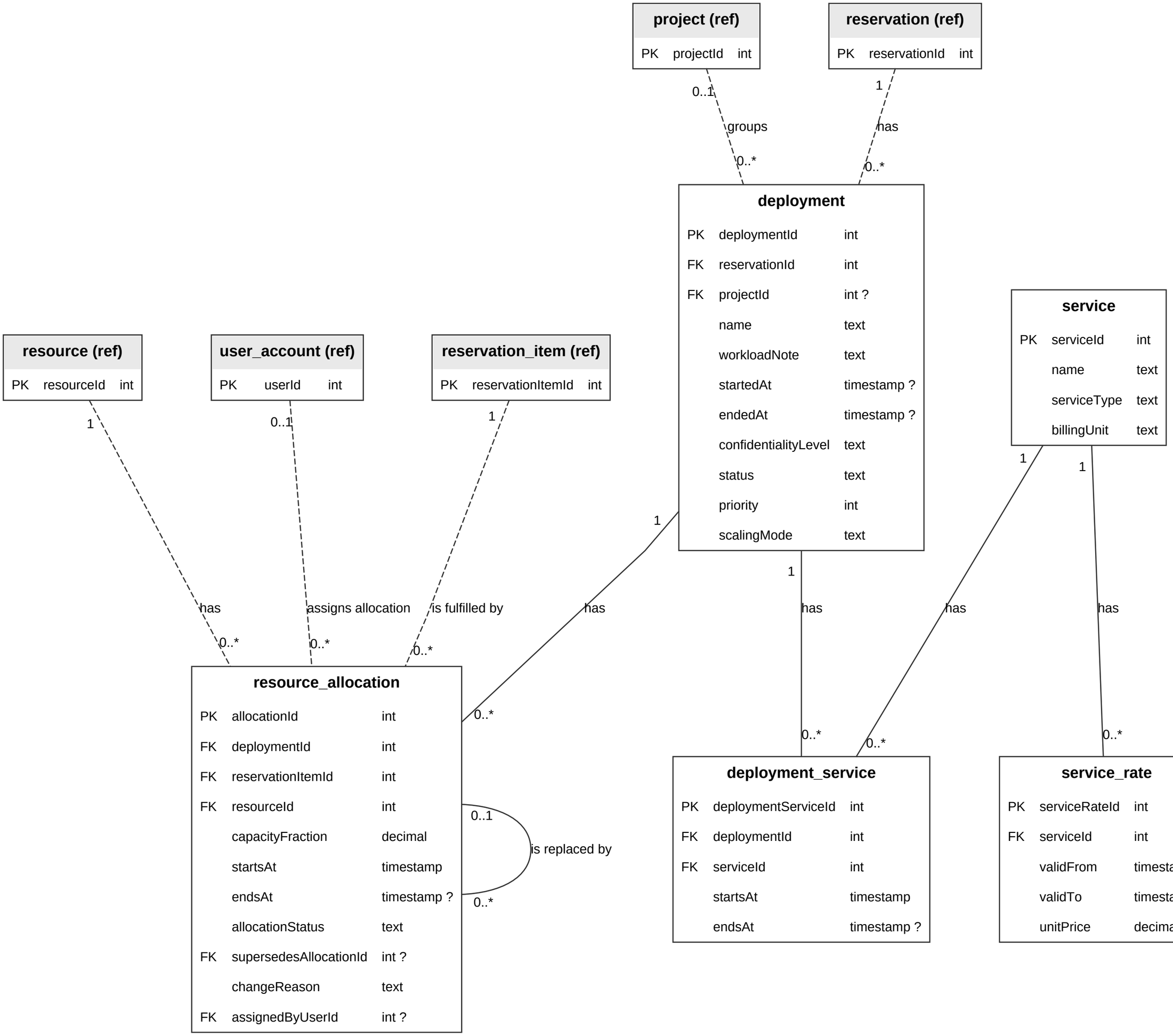
Uniqueness, time intervals, allocation limits and billing rules: analysis report R1-R9.

Each data center belongs to one zone, and each zone belongs to one region.

Resource-state intervals for the same resource cannot overlap; resource capacities are scalar.

Deployments and services - ER diagram

Member C: 5 tables; shaded boxes reference another domain

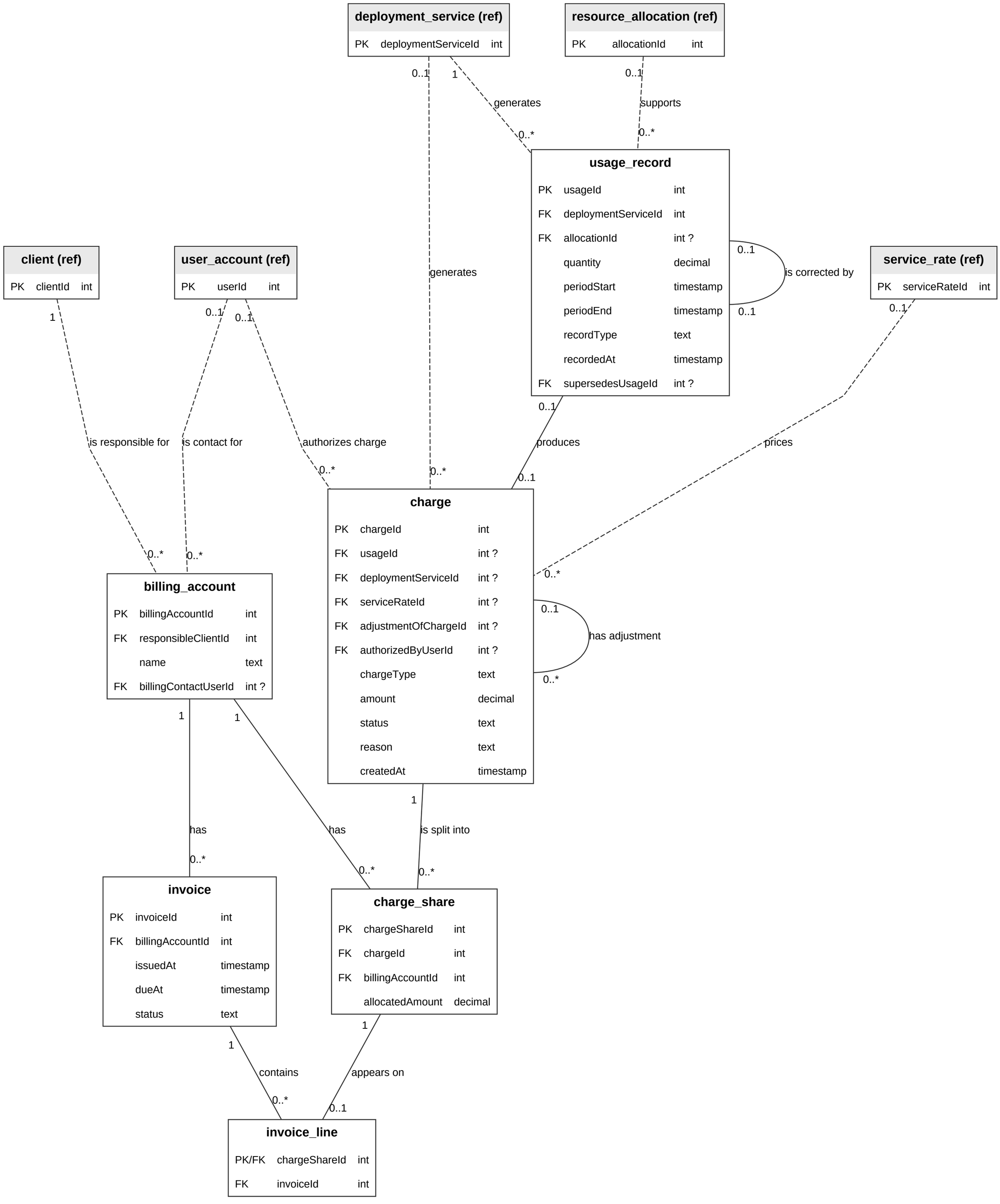


Assumptions and notation

PK = primary key; FK = foreign key; ? = nullable. Unmarked attributes are required.
Endpoint cardinalities: 1 = exactly one; 0..1 = zero or one; 0..* = zero or more.
Each labelled line represents its child foreign key. Reference boxes repeat existing entities.
Uniqueness, time intervals, allocation limits and billing rules: analysis report R1-R9.
Allocations must match the reservation item and deployment; overlapping fractions total at most one.
Replacement allocations retain the predecessor; rates for the same service cannot overlap.

Metering and billing - ER diagram

Member D: 6 tables; shaded boxes reference another domain



Assumptions and notation

PK = primary key; FK = foreign key; ? = nullable. Unmarked attributes are required.
Endpoint cardinalities: 1 = exactly one; 0..1 = zero or one; 0..* = zero or more.
Each labelled line represents its child foreign key. Reference boxes repeat existing entities.
Uniqueness, time intervals, allocation limits and billing rules: analysis report R1-R9.
Each charge has exactly one context: usage, direct service, or adjustment of an earlier charge.
Final payer shares sum to the charge; an invoiced share belongs to the invoice account and is billed once.